ASSESSMENT TOOL -1 (High)

COURSE CODE : CSA 0311

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COURSE NAME : Data structures

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CO2 - ANALYTICAL PROBLEM SOLVING

1. Find the difference between the minimum and Maximum values in a single linked list.

Algorithm :

1. Create a single linked list
2. Initialize min and max with the first node.
3. Traverse the list
4. update min and max
5. Print max - min

code :

```
#include <stdio.h>
#include <stdlib.h>

Struct Node {
    int data ;
    Struct Node * next ;
};

int main() {
  int arr[100], n, i ;
```

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```
Struct Node *head = NULL, *temp, *newnode;

Printf (" Enter the number of nodes : ");

Scanf ("%d", &n);

for(i=0; i<n; i++)

{
  newnode = (struct Node *) malloc ( sizeof (
  struct Node));

  Printf (" Enter data : ");
  Scanf ("%d", & newnode --> data);

  newnode --> next = NULL;

  if (head == NULL)
  {
     head = newnode;
  }
  else
  {
     temp = head;
     while (temp --> next != NULL)
          temp = temp --> next;
```

```
temp --> next = newnode ;
  }
}

temp = head ;
int max = temp --> data , min = temp -> data ;

while ( temp ! = NULL)
{
    if ( temp --> data > max)
           max = temp -> data ;
    if ( temp -> data < min )
           min = temp --> next ;
  }

  Printf (" Maximum = %d\n", max) ;
  Printf (" Minimum = %d\n", min) ;
  Printf (" Difference = %d\n", max - min) ;

  return 0 ;

}
```

Sample input

Enter the number of nodes : 6

Enter data = 34

77

22

55

66

11

Output :

Maximum = 77

Minimum = 11

Difference : 66

Q2. Display the Elements of a double linked list in Forward and Reverse order and find the average.

Algorithm :

1. create a doubly linked list
2. Display nodes from head to tail and from tail to head.
3. Calculate the Sum of all Elements
4. Find the average using Sum / n.

code :

```
#include <stdio.h>
# include < stdlib.h>

Struct Node {
    int data ;
    Struct Node * Prev * next ;
};

int main() {
    int n, i, sum = 0 ;
    Printf (" Enter number of Element : ");
    Scanf ("%d", &n);

    Struct Node * head = NULL . * tail = NULL
        * newnode ;

    for (i = 0 ; i<n ; i++) {
        newnode = (Struct Node *) malloc (siz
(struct Node)) ;
        Printf (" Enter Element : ");
        Scanf ("%d" , & newnode --> data )

        Sum + = newnode --> data ;
        newnode --> next = NULL ;
        newnode --> Prev = tail ;
```

```
if (head == NULL)
    head = newnode ;
else
    tail --> next = newnode ;

tail = newnode ;
}

printf ("forward :") ;
Struct Node * temp = head ;
while (temp != NULL) {
    Printf ("%d ", temp --> data) ;
    temp = temp --> next ;
}

Printf ("\n reverse : ") ;
temp = tail ;
while ( temp != NULL) {
    Printf ("%d ", temp --> data) ;
    temp = temp --> prev ;
}
Printf ( "\n Average = %.2f ", (float) sum/n) ;
return 0 ;
}
```

Input :

Enter number of elements : 5

Enter elements : 10 20 30 40 50

Output :

Forward = 10 20 30 40 50

Backward = 50 40 30 20 10

Average = 30.00